Native D18 cadence-32 input and capacity preflight

Status, 2026-10-26:

- `INPUT_CUSTODY_PASS`
- `CAPACITY_GATE_FAIL`
- `WORKER_NOT_LAUNCHED`

This is a preflight record for the frozen protocol, not a relation search or a factor-base ranking. `selected_best_total_runtime` remains `null`.

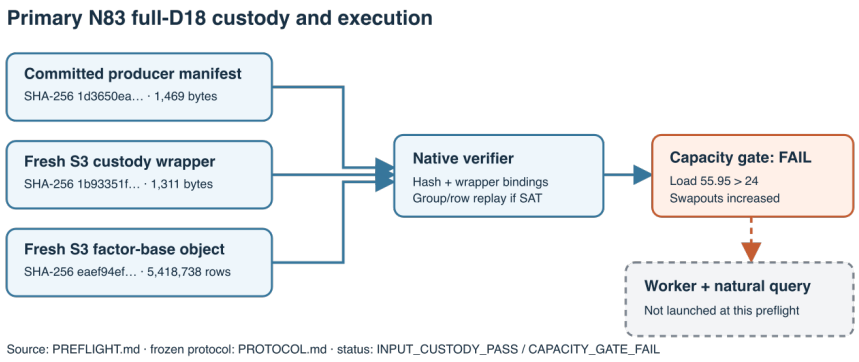


Figure 1: Input custody and execution gates

Exact input and source

The primary curve is Koblitz `a=0` over GF(2^83). Its subgroup order is:

`241785163923079621668568 9`

Its exact curve ID is:

`icv1-f2m83-tm6151469093347-debefd74`

This preflight used the clean source commit on PR #2649:

`1fde65e5e5165dc23de03b555913d8d15f957cce`

Artifact	Bytes	Local SHA-256 prefix
Committed producer manifest	1,469	`1d3650ea…`
S3 custody wrapper	1,311	`1b93351f…`
S3 compressed full-D18 object	160,747,529	`eaef94ef…`

Full local SHA-256 values, in table order:

```
1d3650eabc4662d8f2597498d7ac92b7cd0a4134ac5f8155aaa49c7af3407dce
1b93351febd47dac4de8c5dd39dbf58564d0057a044054baf448d978f6c0d616
eaef94ef45b3d2d4cd2ec2f40c1173b4cdb2a437b7e618a858c73fb1caf3902b
```

Both S3 objects were freshly fetched into scratch from this bucket:

```
crypto-autoresearcher
```

The AWS handoff gives their complete keys. The retained provider GET receipts are wrapper and object. Their SHA-256 values follow:

```
3cc440bf5e5eb6a943dd534f6af5cb48f49b4b3d3ebf87eed62a701f96a49172
1578b28a45f0a4f892bbb013ea10fcdd126fd543105d7b2084bac6e74610835a
```

Their base64 `ChecksumSHA256` fields decode to the local hashes above; both report `ChecksumType=FULL_OBJECT` and AES256 at rest. The wrapper binds the producer manifest, curve, policy, object key, 32,643 columns and 5,418,738 point rows. The native verifier now checks these distinct manifests and bindings before replay.

The release worker and verifier SHA-256 values are, respectively:

```
afd2fe3870f758d96dd97f1982c85d65d6ebb210659133dc93b1d3f5fda1146b
5f7679ee4b001438bdd27de49f8a4935f9e9803bf00a56e0045f502d1a7e1ab9
```

Both were built from the clean commit above:

```
cargo build --release \
  --example koblitz_n83_factor_base_export \
  --example n83_d18_native_relation_replay \
  --target-dir /private/tmp/n83-evidence-cargo-target
```

Validation and capacity

The committed patch passed the focused verifier tests (3), historical N83 suite (63, plus 70 subtests), and boundary suite (52, plus 9 subtests). The full release library rerun with loopback access and four test threads passed 4,168 tests, with 117 ignored and zero failed. The retained compressed log is 67,218 bytes. Its compressed and uncompressed SHA-256 values are, respectively:

```
c87c54d070460b5c56d77d027240ae0f4d9ae610889fa0be18c5ebb541860cb3
b129cdeb7e2b88cecd80357b02b6f17381a0d308532869d9dae59c5c68328d8a
```

An earlier default-sandbox full-suite attempt failed 14 existing tests under restricted loopback and host contention; it is not a passing gate. Focused Clippy, targeted rustfmt and diff checks passed; Clippy emitted the existing unknown-lint warning from the repository configuration.

The protocol requires one-minute host load at most 24, no swapouts during the run, and at least 2 GiB scratch free. The shared Mac had 86 GiB scratch free at

12:52 PDT, but sampled one-minute load was **74.19** at 12:51, **63.05** at 12:52, **66.80** at 12:55 and **55.95** at 12:56. The `vm_stat` cumulative swapout count rose from 299,564 at 12:51 to 353,036 at 12:56. These are sparse host observations, not a continuous utilization trace. This capacity state fails the frozen launch gate. The worker was not started, and neither a positive nor a natural outcome exists for this checkpoint.

One connection attempt to `awsspot` timed out during SSH banner exchange; the configured `runpod` and `runpod-isolated-blush-stork` endpoints each refused the connection. The AWS spot lifecycle controller reported a ready compute instance but `ssh_endpoint_ready=false`; its bounded wake request did not make the endpoint ready within 30 seconds. The hosted AWS principal's exact-object read grant remains unapplied. GitHub Actions for PR #2649 failed before runner assignment with the account billing/spending-limit annotation recorded in the PR comment. None of these are solver observations.

Next gate

On a reachable authorized host, check the source and fresh full-object S3 GETs, rebuild and hash both release binaries, and record a host preflight. Only then launch one 300-second internal-cap constructed control. A passing native group/row replay is required before one 2,400-second natural query. Independent assignment replay, useful-rank closure, full-width logs, unseen-target descent, fully charged cold one-target cost and matched rho remain necessary before selecting a factor base by total runtime. The existing problem record tracks those unresolved gates.

The visual source is CUSTODY.svg. Its PNG was rendered from that SVG at 1,406 × 609 pixels with `rsvg-convert` and tagged 150 dpi with `sips`. The report PDF was built from this Markdown and PNG with Pandoc and Tectonic.